

VenusFold Improves Biomolecular Interface Prediction through a Refined Training Strategy

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Understanding biomolecular interactions requires accurate prediction of the three-dimensional structures and interfaces formed by interacting proteins, peptides, nucleic acids, and small molecules. Predicting these interactions requires models capable of jointly generating accurate all-atom structures of interacting molecules while capturing the complex geometries of their interfaces. We present VenusFold, an all-atom biomolecular structure prediction model designed for accurate complex structure prediction with improved diffusion training strategies. VenusFold is trained using minimum signal-to-noise ratio (Min-SNR) diffusion-loss weighting and the Momentum Orthogonalized by Newton-Schulz (Muon) optimizer to balance loss contributions and improve the training efficiency. On the FoldBench, VenusFold achieved state-of-the-art protein-ligand performance among the baseline models, with an LDDT-PLI of 0.805 and a docking success rate of 69.59%. VenusFold also achieved the strongest rank-selected results for protein-protein, protein-peptide, protein-DNA, protein-RNA and protein-monomer prediction. These results position VenusFold as a broadly capable all-atom predictor, with strengths in interface prediction across proteins, peptides, ligands and nucleic-acids, providing essential structural information for studying molecular interaction and supporting structure-based drug design.

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Repository: <https://github.com/ai4protein/VenusFold>

1 Introduction

Biological function depends on both the structures adopted by biomolecules and the interactions they form with molecular partners. Predicting these structures directly from molecular composition has therefore been a longstanding objective of computational biology, with applications ranging from the interpretation of molecular mechanisms to the discovery of therapeutic molecules. Recent deep-learning methods have transformed protein structure prediction and are now extending this capability from individual protein chains to heterogeneous complexes containing proteins, nucleic acids, small molecules, ions and modified residues. Joint all-atom prediction offers a unified route for modeling these systems without separating folding, docking and atomic refinement into independent stages.

AlphaFold demonstrated that a generative all-atom model can predict a broad range of biomolecular complexes within a shared architectural framework (Abramson et al., 2024). Subsequent models, including Chai-1 (Chai Discovery Team, 2024), Boltz-2 (Passaro et al., 2025), Protenix v2 (Zhang et al., 2026), RoseTTAFold3 (RosettaCommons, 2026) and OpenFold3 Preview 2 (OpenFold3 Team, 2026), have further advanced and broadened this co-folding paradigm. Nevertheless, support for diverse molecular types does not necessarily translate into uniformly accurate prediction across them. Models that accurately recover protein folds may still struggle to resolve protein-protein interfaces, protein-nucleic-acid recognition or small-molecule binding poses. The remaining challenge is therefore not merely to accommodate more molecular components, but to learn reliable global structures and local interface geometries across chemically distinct systems within a single model.

Achieving such breadth presents a coupled optimization problem. The representation trunk and coordinate generator must improve together across polymer and interface types that differ substantially in data frequency and geometric difficulty. Diffusion training adds a second source of imbalance because noise levels induce denoising tasks of different difficulty and gradient scale. Increasing model capacity can improve representation power, but it also makes optimization of the large hidden matrix parameters more expensive. Finally, inference produces multiple candidate structures, so practical quality depends on both coordinate generation and confidence-based candidate selection. We focus on improving this training process without changing the overall all-atom architecture or adding external geometric constraints at inference time.

Here we present VenusFold, a scaled all-atom biomolecular structure predictor whose overall architecture broadly follows AlphaFold 3 (Abramson et al., 2024). MSA and template features condition the learned single and pair representations, and an atom-level diffusion model generates the joint all-atom structure of the complex. Within this architecture, VenusFold modifies two complementary components of the training recipe. First, VenusFold applies minimum signal-to-noise ratio (Min-SNR) weighting on top of EDM preconditioning (Karras et al., 2022)(Hang et al., 2023). Min-SNR limits the relative contribution of high-SNR diffusion states and is intended to reduce optimization conflicts among noise regimes. Second, VenusFold introduces the Momentum Orthogonalized by Newton–Schulz (Muon) optimizer for matrix-valued parameters (Jordan et al., 2024). Together, Min-SNR and Muon address complementary parts of the training problem. The question considered here is whether diffusion weighting and Muon optimization can improve an all-atom predictor while preserving the established architecture and inference procedure.

We assess the optimizer design in a controlled short-horizon experiment and evaluate the complete trained model against eight other structure-prediction models on the same FoldBench targets. The evaluation uses aligned sampling budgets, MSA inputs and template policies across compatible models. Under this protocol, VenusFold achieves the strongest rank-selected performance for protein–ligand, protein–protein, protein–DNA, protein–RNA and protein–monomer structure prediction, while tying the highest protein–peptide docking success rate. These results demonstrate broad benchmark capability across monomers and diverse interaction classes, with particularly strong performance for protein–ligand and protein–nucleic-acid complexes.

2 Method

2.1 Conditioned diffusion for all-atom protein structure generation

Let a biomolecular complex contain one or more polymer chains with sequences $S = \{s_k\}_{k=1}^K$ and a molecular specification m comprising its stoichiometry, residue and atom identities, and covalent connectivity. The structure-prediction problem is to infer the joint Cartesian coordinates $X \in \mathbb{R}^{N \times 3}$ of its N resolved atoms. Optional multiple-sequence alignments (MSAs) and structural templates are included with (S, m) in the conditioning information c . Modern structure prediction models the conditional distribution

$$p_{\theta}(X \mid c), \quad c = \{S, m, \text{MSA}, \text{template}\}, \quad (1)$$

and samples joint all-atom structures together with per-atom, pairwise, and complex-level confidence estimates. This formulation supports complexes containing proteins, DNA, RNA, small molecules in a shared representation.

The overall architecture of VenusFold broadly follows AlphaFold 3 (Abramson et al., 2024). Atom- and token-level input features are first embedded into single and pair representations. MSA and template modules provide evolutionary and structural conditioning, after which the Pairformer stack repeatedly updates the single and pair states. Conditioned on these representations, an atom-level diffusion module transforms noisy coordinates into a coherent structure for the entire complex. A distogram head provides an auxiliary geometric objective, and a separate confidence head estimates the quality of the sampled structure. We also replace rectified linear unit (ReLU) activations with the sigmoid-weighted linear unit (SiLU), $\text{SiLU}(z) = z \text{sigmoid}(z)$, whose smooth negative branch reduces the risk of persistently inactive neurons caused by the zero-gradient negative regime of ReLU (Elfwing et al., 2018).

Coordinate generation is performed with a conditional diffusion model. During training, Gaussian noise at level σ is added to a reference all-atom structure X_0 to obtain X_σ . The denoising network predicts $\hat{X}_0 = f_\theta(X_\sigma, \sigma, c)$, thereby learning structural constraints that are compatible with both the molecular sequence information and the encoded complex context. At inference, the model starts from noisy atom coordinates and iteratively denoises them to generate a joint all-atom complex structure.

2.2 Fine-grained Min-SNR diffusion-loss weighting

Diffusion training jointly optimizes denoising tasks across a broad range of noise levels. These tasks differ in difficulty and gradient scale, so high-signal-to-noise ratio (SNR) samples can contribute disproportionately and dominate a minibatch. To allocate optimization effort more evenly across noise regimes, we apply a fine-grained, sample-specific minimum-SNR (Min-SNR) correction to the diffusion loss.

Let X_0 be a reference all-atom structure, X_σ its randomly rotated, translated, and Gaussian-corrupted version at noise level σ , and $\hat{X}_0 = f_\theta(X_\sigma, \sigma, c)$ the denoised prediction conditioned on the encoded input c . Noise levels are sampled from $\log \sigma \sim \mathcal{N}(-1.2, 1.5^2)$ with $\sigma_{\text{data}} = 16$. We start from the EDM preconditioning weight (Karras et al., 2022)

$$w_{\text{EDM}}(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{\sigma_{\text{data}}^2 \sigma^2} \quad (2)$$

and apply Min-SNR weighting (Hang et al., 2023) separately to each diffusion sample:

$$w(\sigma) = w_{\text{EDM}}(\sigma) \frac{\min(\text{SNR}(\sigma), \gamma)}{\text{SNR}(\sigma)}, \quad \text{SNR}(\sigma) = \left(\frac{\sigma_{\text{data}}}{\sigma} \right)^2, \quad (3)$$

where $\gamma = 5$. For samples with $\text{SNR}(\sigma) > \gamma$, the factor $\gamma / \text{SNR}(\sigma)$ reduces their relative contribution; samples below the threshold retain their EDM weight. This cap keeps low-noise, high-SNR examples from overwhelming the gradient and directs a larger fraction of each update toward less well-optimized noise regimes. The resulting balance reduces competition among denoising tasks and improves the efficiency of diffusion training across noise scales. We do not renormalize the batch mean of these weights, and Min-SNR is used only during training.

The initial-stage objective combines the Min-SNR-weighted coordinate loss with geometric and confidence objectives:

$$\mathcal{L}_{\text{initial}} = 4 (\mathcal{L}_{\text{MSE}}^{\text{MinSNR}} + \mathcal{L}_{\text{smooth-LDDT}}) + 0.03 \mathcal{L}_{\text{distogram}} + 10^{-4} \mathcal{L}_{\text{confidence}}. \quad (4)$$

The aligned coordinate MSE upweights DNA and RNA atoms by 5 and ligand atoms by 10 relative to protein atoms. The distogram term is a cross-entropy objective over representative-atom distances. The confidence term comprises pLDDT, predicted distance error, and resolved-atom losses during these stages; the PAE component is reserved for confidence-head training.

2.3 Momentum Orthogonalized by Newton–Schulz optimization for matrix-valued parameters

We introduce the Momentum Orthogonalized by Newton–Schulz (Muon) optimizer to improve the optimization of matrix-valued parameters (Jordan et al., 2024). For a matrix-shaped parameter gradient G_t , Muon first forms a Nesterov momentum update

$$B_t = \mu B_{t-1} + G_t, \quad M_t = G_t + \mu B_t, \quad (5)$$

with $\mu = 0.95$. Muon treats M_t as a matrix-valued update and replaces its singular values with a common scale. If $M_t = U \Sigma V^\top$, the resulting target direction is the semi-orthogonal polar factor

$$\begin{aligned} \mathcal{O}(M_t) &= UV^\top = \arg \min_{Q \in \mathcal{S}_{m,n}} \|Q - M_t\|_F, \\ \mathcal{S}_{m,n} &= \begin{cases} \{Q : QQ^\top = I_m\}, & m \leq n, \\ \{Q : Q^\top Q = I_n\}, & m > n. \end{cases} \end{aligned} \quad (6)$$

This preserves the left and right singular directions of the momentum update while mapping its nonzero singular values toward a common scale, preventing a few high-gain modes from dominating solely because of their magnitude.

Computing an SVD for every parameter matrix would be expensive. Convolutional or higher-order tensors are therefore flattened along all but their first dimension, and wide matrices are temporarily transposed. After Frobenius-norm normalization, we approximate the polar factor using five Newton–Schulz iterations composed only of matrix multiplications,

$$X_{k+1} = aX_k + (bA_k + cA_k^2)X_k, \quad A_k = X_kX_k^\top, \quad (7)$$

with $a = 3.4445$, $b = -4.7750$, and $c = 2.0315$. After restoring the original orientation, the update for an $m \times n$ flattened matrix is

$$\theta_{t+1} = \theta_t - \eta \max\left(1, \sqrt{m/n}\right) \hat{\mathcal{O}}(M_t), \quad (8)$$

where $\hat{\mathcal{O}}$ denotes the finite-step Newton–Schulz approximation; the aspect-ratio factor keeps update scales comparable across rectangular matrices.

Muon uses a peak learning rate of 10^{-3} , five Newton–Schulz steps, Nesterov momentum 0.95, and zero weight decay. This produces a hybrid optimizer configuration in VenusFold: Muon is applied only to a subset of matrix-valued parameters, while all remaining parameters are optimized with AdamW (Loshchilov and Hutter, 2019), using a peak learning rate of 10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, and weight decay 10^{-8} .

2.4 Data Processing

The training corpus integrated three complementary sources of experimental and predicted protein-complex structures. These comprised the Protein Data Bank (PDB) Berman et al. (2000), an AFDB-derived quaternary-structure distillation dataset Han et al. (2026), and Teddymer Didi et al. (2026).

PDB dataset. We used the training set released with Protenix-v2, which contains PDB structures released on or before 30 September 2021 Zhang et al. (2026). Protein polymer sequences were extracted from the corresponding mmCIF files and deduplicated at 100% sequence identity before multiple-sequence alignment (MSA) construction.

Each unique sequence was searched independently using JackHMMER Eddy (2011). Following the AlphaFold 3 data- processing configuration Abramson et al. (2024), JackHMMER was run for one iteration (`-N 1`). Both the reporting and inclusion E-value thresholds were set to 1×10^{-4} (`-E 0.0001` and `-incE 0.0001`). The acceleration-filter thresholds were `-F1 0.0005`, `-F2 0.00005`, and `-F3 0.000005`.

Searches were performed against UniProt (2021_04), UniRef90 (2022_05), MGnify (2022_05), and the first non-consensus sequence set of the Big Fantastic Database (BFD). These database releases follow those used by the AlphaFold 3 data pipeline Abramson et al. (2024). UniProt hits were used to construct the pairing MSA, whereas UniRef90, MGnify, and BFD supplied the non-pairing MSA. The latter sources were capped at 10,000, 5,000, and 5,000 sequences, respectively. The database searches were distributed across a computing pool comprising 20,480 CPU cores. The resulting homologous sequences were converted into model-ready A3M alignments.

AFDB-derived quaternary-structure distillation dataset. We constructed a quaternary-structure distillation dataset from the homomeric and heteromeric complex predictions released through the proteome-scale expansion of the AlphaFold Database (AFDB) Han et al. (2026). The source study evaluated more than 31 million candidate complexes and released approximately 1.8 million high-confidence or tentatively high-confidence structures.

Training on the complete release would introduce substantial structural redundancy and unnecessary computational cost because recurrent complex architectures are strongly over-represented in this collection. We therefore retained a structurally diverse set of representative complexes. Models were first screened using the confidence criteria defined in the source study: $ipSAE_{\min} \geq 0.6$, mean pLDDT ≥ 70 , and no more than

10 backbone clashes [Han et al. \(2026\)](#). Here, $ipSAE_{\min}$ denotes the smaller of the two directional ipSAE values calculated for the two chain orderings.

The retained complexes were clustered using Foldseek Multimercluster with the configuration reported for the AFDB expansion and Teddymer datasets [Han et al. \(2026\)](#); [Didi et al. \(2026\)](#). Specifically, clustering required 60% 3Di- sequence coverage, 30% interface similarity, and 70% chain similarity (`-c 0.6`, `-interface-lddt-threshold 0.3`, `-chain-tm-threshold 0.7`, and `-cov-mode 0`). One representative structure was selected from each cluster, reducing the highly redundant release to 195,670 candidate representatives. File-integrity verification, protein-chain extraction, non-standard-residue normalization, structural validation, and exact-sequence deduplication subsequently removed 13 unavailable, invalid, or redundant representatives. The final distillation dataset therefore contained 195,657 unique complex structures.

Teddymer dataset. Teddymer is a large-scale synthetic dimer dataset constructed from predicted monomer structures in AFDB [Didi et al. \(2026\)](#). Structural-domain annotations from the Encyclopedia of Domains (TED) were used to partition multidomain AFDB monomers into individual domains. Spatially interacting domain pairs were then represented as separate chains of synthetic protein complexes. The resulting assemblies were filtered using interface-geometry and prediction-confidence criteria and structurally clustered to reduce redundancy.

For the present study, we applied a unified workflow comprising structure-file validation, protein-chain extraction, residue normalization, protein-sequence extraction, exact-sequence deduplication, MSA generation, and template search. The JackHMMER parameters and sequence databases were identical to those described for the PDB dataset. After processing, the Teddymer dataset associated 609,271 unique protein sequences with their corresponding synthetic dimer structures.

2.5 Training configuration

Training used the hybrid optimizer described above. Matrix-valued parameters in the Pairformer and MSA modules used Muon with a peak learning rate of 10^{-3} , Nesterov momentum 0.95, five Newton-Schulz iterations, and zero weight decay. All remaining trainable parameters used the AdamW fallback with the same peak learning rate, $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$, and weight decay 10^{-8} . The AlphaFold 3-style schedule applied linear warmup followed by a multiplicative learning-rate decay of 0.95 every 50,000 updates. Gradients were clipped to a global norm of 10.

For diffusion training, Min-SNR weighting was applied on top of EDM preconditioning as defined in [Equation \(3\)](#), with $\gamma = 5$. The resulting weights were not renormalized by their batch mean. This weighting modified only the diffusion coordinate objective; auxiliary losses retained their configured weights, and Min-SNR was not used during inference.

Training ran on 10 nodes with eight 80-GB NVIDIA A800 GPUs per node using bfloat16 mixed precision and TF32. Model weights were tracked with an EMA decay of 0.999. Efficient fusion, fast layer normalization, and cuEquivariance triangle kernels were enabled; diffusion computations used chunks of four samples. The model used four recycling cycles during training, and all reported runs used seed 42.

3 Results

3.1 Performance across FoldBench tasks

We evaluated VenusFold against eight other structure-prediction models across nine FoldBench task categories: protein, RNA, and DNA monomers and six types of complexes (protein-protein, antibody-antigen, protein-peptide, protein-DNA, protein-RNA, and protein-ligand) ([Xu et al., 2026](#)). The comparison included AlphaFold 3 ([Abramson et al., 2024](#)), Protenix v2 ([Zhang et al., 2026](#)), OpenDDE ([Aureka AI OpenDDE Project, 2026](#)), ESMFold2 ([Candido et al., 2026](#)), Chai-1 ([Chai Discovery Team, 2024](#)), RoseTTAFold3 ([RosettaCommons, 2026](#)), Boltz-2 ([Passaro et al., 2025](#)), and OpenFold3 ([OpenFold3 Team, 2026](#)). Metric means

and success rates are computed by the FoldBench evaluator on this shared target set. The reported local, interface, and global metrics follow the IDDT (Mariani et al., 2013), DockQ v2 (Mirabello and Wallner, 2024), and TM-score (Zhang and Skolnick, 2004), respectively. Protein–ligand joint success requires ligand RMSD at most 2 Å and IDDT-PLI at least 0.8. Under best-of-five selection, VenusFold achieved an IDDT-PLI of 0.800, compared with 0.797 for AlphaFold 3, the strongest competing result. Joint protein–ligand success was 69.78%, exceeding Boltz-2 by 1.18 percentage points and AlphaFold 3 by 2.41 points. Protein-pocket IDDT-LP was 0.915.

The advantage extended beyond small-molecule docking. VenusFold obtained 75.38% protein–protein docking success, 85.53% protein–DNA docking success, and 75.38% protein–RNA docking success. The corresponding strongest competing results were 73.86%, 80.50%, and 73.85%, respectively. VenusFold reached 48.48% antibody–antigen docking success and an antibody–antigen interface IDDT of 0.835, while protein–peptide docking success was 88.37%.

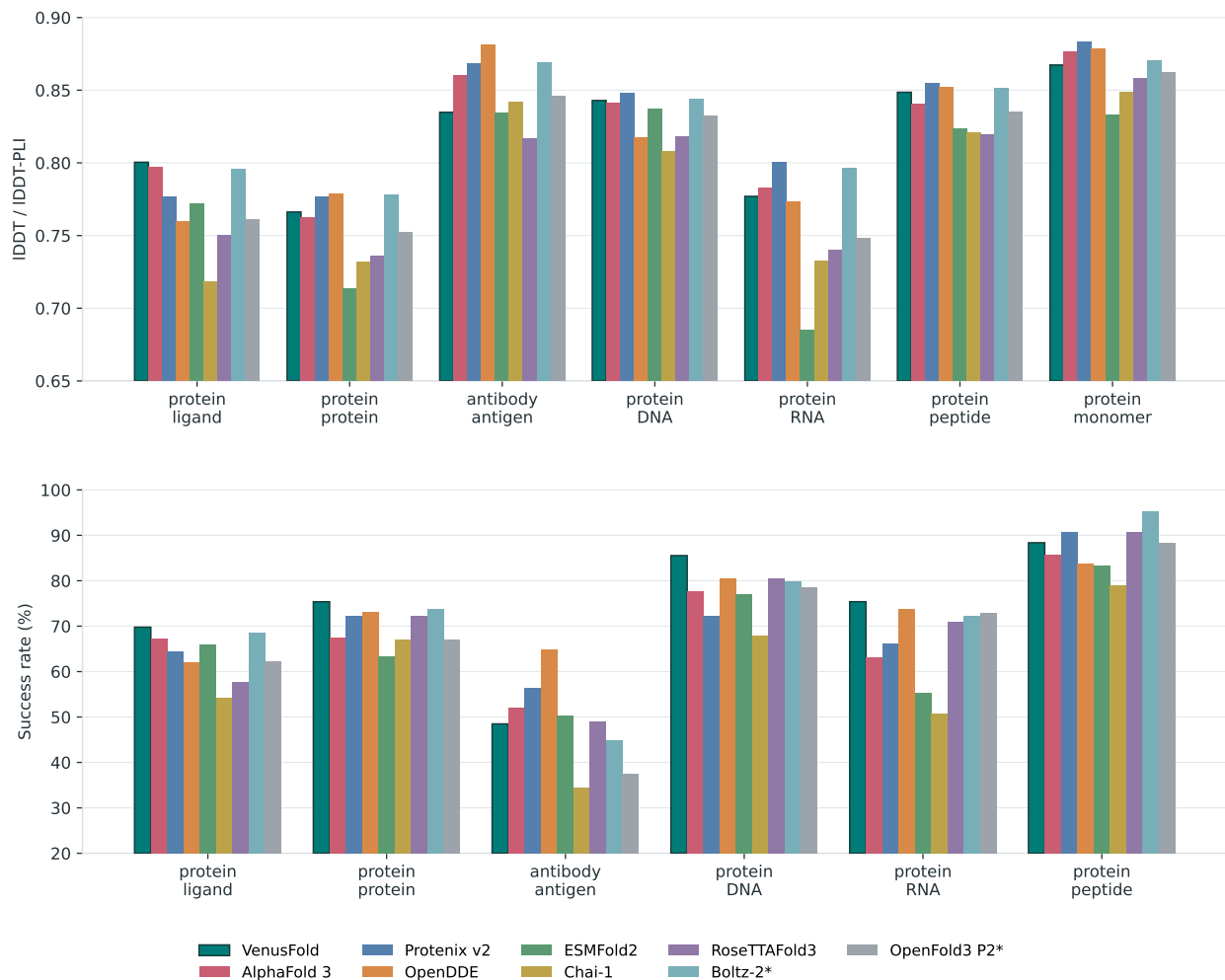


Figure 1 FoldBench evaluation across all nine models. Both panels use oracle best-of-five predictions for each reported metric. DNA and RNA monomers are not included. Boltz-2 and OpenFold3 Preview 2 are post-cutoff references. ESMFold2 is sequence-only and was run without MSA input.

Aggregate improvements were also visible at the structure level (Figure 2). Across 8GFD, 8PGM, and 8R6J, the confidence-ranked VenusFold poses had ligand RMSDs of 0.98, 1.36, and 1.12 Å, respectively, compared with 18.02, 9.02, and 10.99 Å for AlphaFold 3. The structures were aligned on the protein before comparing ligand placement. These cases were selected after inspecting performance differences and are explanatory rather than independent statistical evidence.

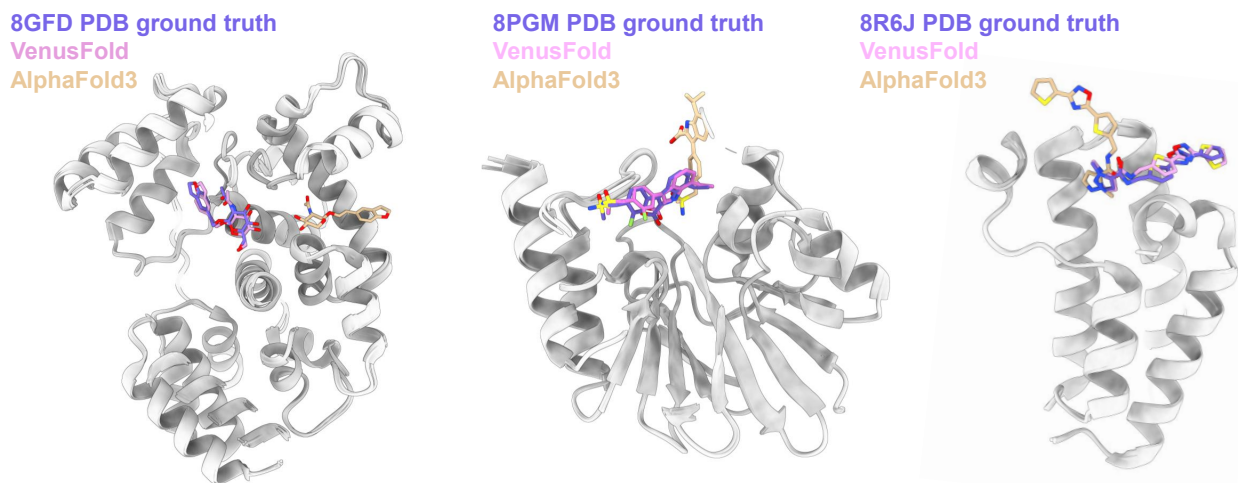


Figure 2 Overlaid confidence-ranked protein–ligand predictions for 8GFD, 8PGM, and 8R6J after alignment on the protein backbone. Colored ligands show the PDB ground truth, VenusFold, and AlphaFold 3 poses as labeled.

3.2 Training-strategy ablation

We conducted a 1,000-step short-horizon comparison between the original AdamW–EDM training recipe (origin) and the VenusFold recipe combining Muon with Min-SNR weighting. Both runs used ReLU activations, seed 42, a peak learning rate of 10^{-3} , 384-token crops, a 256-token input cap, and an effective batch size of 24. We evaluated both configurations every 50 optimizer steps on the same validation set and report confidence-ranked IDDT together with validation loss.

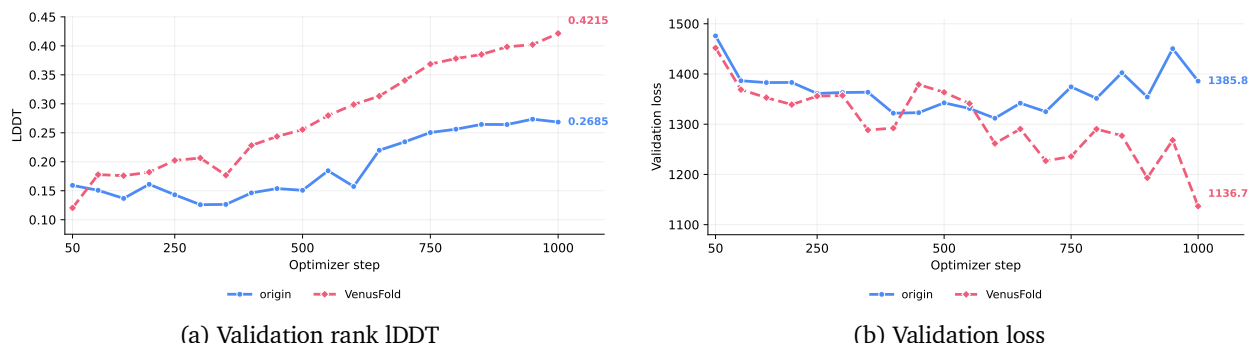


Figure 3 Short-horizon validation trajectories for the original AdamW–EDM recipe (origin) and the VenusFold Muon–Min-SNR recipe. (a) Rank-selected IDDT, where higher is better. (b) Validation loss, where lower is better. Both runs use seed 42 and are evaluated at 50-step intervals.

The two configurations were similar during the earliest evaluations, after which their structural-accuracy trajectories separated. From approximately 400 steps onward, VenusFold showed a sustained increase in rank IDDT, whereas the origin curve improved more slowly and began to plateau. At step 1,000 (Figure 3a), rank IDDT reached 0.4215 for VenusFold and 0.2685 for origin, an absolute gain of 0.1530 and a relative gain of 57.0%.

The validation-loss trajectory was noisier but showed a consistent downward trend for VenusFold in the latter half of training. At step 1,000 (Figure 3b), the reported validation loss was 1136.7, compared with 1385.8 for origin, a reduction of 249.1 (18.0%). Because Min-SNR changes the weight assigned to diffusion noise levels, these loss values do not represent an identical scalar objective across the two configurations. The rank-IDDT improvement therefore provides the more direct evidence of improved structural prediction in this experiment. This short-horizon comparison uses one training seed and should be interpreted as evidence of early optimization behavior rather than a converged or multi-seed estimate of the final performance gain.

4 Conclusion and Discussion

The results of VenusFold demonstrate that the performance of all-atom biomolecular structure prediction models depends not only on architectural design but also on effective training strategies. By incorporating Min-SNR loss weighting and Muon optimization, VenusFold achieves improved training process and enhanced predictive performance, highlighting the importance of diffusion objectives and optimization algorithms in molecular modeling. Evaluation on FoldBench further shows that VenusFold provides substantial improvements across diverse molecular interaction types, including protein–ligand, protein–protein, protein–peptide, and protein–nucleic-acid complexes. These results suggest that improved training recipes can enhance the general capability of all-atom structure prediction models across different interaction scenarios. However, the improvement on antibody–antigen complexes remains limited, indicating that certain interaction types may require further methodological development.

Despite recent advances, current all-atom structure prediction models still have considerable room for improvement. In our evaluation, antibody–antigen complexes exhibit distinct performance characteristics compared with other interaction types, with VenusFold showing limited improvement. Interestingly, similar trends have been observed in other recent structure prediction studies, such as OpenDDE (Aureka AI OpenDDE Project, 2026), where improvements are concentrated on antibody–antigen prediction, indicating that antibody–antigen interaction classes may require specialized approaches. Our future work will investigate the scaling behavior of all-atom structure prediction models by expanding model size, representation capacity, and training data, aiming to determine whether increased scale can provide consistent improvements across diverse molecular interaction types and address current limitations.

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